Optimal Additive Integer Masks for One-Guess Vulnerability under an Absolute-Moment Budget

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November 11, 2026

Abstract

We study the average one-guess success probability of an adversary who observes the ordinary integer sum of a finite-support secret and an independent integer-valued mask, subject to an absolute-moment budget on the mask. For a uniform secret on an interval the exact optimum and an explicit optimiser are obtained. For a biased bit the masks that achieve no increase over the prior baseline are characterised, their minimal cost is determined, the initial segment of the cost–vulnerability frontier is obtained, and two further affine pieces are certified by dual multipliers for biases at most the golden-ratio conjugate. The remaining intermediate frontier is left open. The results give exact design guidance for additive integer noise under the average one-guess metric; they do not constitute a cryptographic masking scheme.

Keywords: Bayes vulnerability, quantitative information flow, additive noise, absolute-moment constraint, geometric distributions.

1 Introduction

Quantitative information flow measures how much an observation helps an adversary guess a secret. The most basic operational criterion is the probability that the adversary's single best guess is correct. This is the Bayes vulnerability $V(X) = \max_x \mathbb{P}(X = x)$ before the observation and the posterior vulnerability

$$V(X \mid S) = \sum_s \max_x \mathbb{P}(X = x, S = s)$$

after observing S [1]. The average conditional min-entropy is $-\log_2 V(X \mid S)$ [2]. Minimising $V(X \mid S)$ is therefore equivalent to maximising average conditional min-entropy. Maximal leakage [4], worst-case posterior confidence, mutual information, and differential privacy answer different threat models and are not studied here.

We consider the channel $S = X + Y$ in which an independent integer-valued mask Y (finite or infinite support) is added to a finite-support integer-valued secret X. The design question is how to choose the law q of Y so as to minimise average one-guess success subject to $\mathbb{E}|Y| \le c$. Absolute-moment costs arise when the practical price of noise is proportional to its magnitude. Zero additional average guessing advantage does not imply independence or perfect secrecy.

Contributions.

1. For a uniform secret on an interval of length m the exact minimal vulnerability and an explicit optimiser are determined (Theorem 4.4).

2. For a biased bit all masks that produce no extra average guessing advantage are characterised, the exact critical cost and the strict gap below it are given, and the exact initial segment is determined (Theorems 5.1–5.5).

3. For biases $1/2 < p \le (\sqrt{5}-1)/2$ two further affine pieces of the frontier are certified by explicit dual multipliers (Theorem 5.6). The general intermediate frontier is stated as an open problem. Figure 1 shows the proved segments (solid) together with finite-support LP upper bounds (dashed); the dashed curves are not certified.

2 Related work

Smith [1] introduced Bayes vulnerability and min-entropy leakage. Dodis, Ostrovsky, Reyzin and Smith [2] defined average min-entropy. Braun, Chatzikokolakis and Palamidessi [3] compared additive and multiplicative one-try leakage. Issa, Wagner and Kamath [4] showed that maximal leakage for finite alphabets equals $\log \sum_s \max_{x:p(x)>0} \mathbb{P}(S = s \mid X = x)$.

Saeidian, Cervia, Oechtering and Skoglund [5] solved a privacy–utility problem under maximal leakage and expected Hamming distortion; the mechanisms here are restricted to be additive and the cost is an absolute moment. Asoodeh et al. [6] studied privacy-constrained guessing probabilities under a different formulation that does not specialise to additive integer masks with an absolute-moment budget. Geng, Ding, Guo and Kumar [8] derived optimal additive noise for differential privacy under ℓ_1 and ℓ_2 costs; the constraint is differential privacy, not Bayes vulnerability.

Side-channel masking literature (e.g., the noisy-leakage model of Prouff and Rivain [7]) addresses a different threat model based on shares. A targeted search of the primary quantitative-information-flow, maximal-leakage, differential-privacy-noise, and guessing-with-noise literature did not locate prior solutions of the exact cost-constrained additive-integer problems studied here. This is an observation about the sources examined, not a proof of novelty.

3 Preliminaries

Let X have finite support and law p, and let Y be independent with law q on $\mathbb{Z}$. Set $S = X + Y$.

Proposition 3.1 (MAP rule)**.** *The optimal one-guess success probability is*

$$V(X \mid S) = \sum_{s \in \mathbb{Z}} \max_x p(x)q(s-x),$$

attained by any $g(s) \in \arg\max_x p(x)q(s-x)$*. In particular* $V(X \mid S) \geq V(X)$*.*

Proof. A strategy g succeeds with probability $\sum_s p(g(s))q(s-g(s))$; maximise each term separately. For any fixed x_0, the sum is at least $p(x_0)$. □

For a fixed secret distribution the optimisation is over all probability laws q on $\mathbb{Z}$ with $\mathbb{E}_q|Y| \leq c$. The optimal value is denoted $V^*(c)$.

4 Uniform interval secret

Let X be uniform on $I_m = \{0, \ldots, m-1\}$. Then

$$V(q) = \frac{1}{m} \sum_s \max_{x \in I_m} q(s-x).$$

Lemma 4.1 (Layer cake)**.** *Let* $h = \|q\|_\infty$ *and* $B_t = \{y : q(y) > t\}$*. Then* $mV(q) = \int_0^h |B_t + I_m|\, dt$*.*

Proof. For each s, $\max_{y \in s - I_m} q(y) = \int_0^h \mathbf{1}[s \in B_t + I_m]\, dt$. Sum over s and apply Tonelli's theorem. □

Lemma 4.2 (One-dimensional sumset)**.** *For nonempty finite* $A \subset \mathbb{Z}$*,* $|A + I_m| \geq |A| + m - 1$*. For* $m \geq 2$*, equality holds if and only if* A *is an interval of integers.*

Proof. $A + I_m \supseteq A \cup (\max A + I_m)$, and the two sets meet only in $\max A$. If A is an interval the sumset is an interval of the stated cardinality. If consecutive elements of A differ by at least 2, a gap point appears in the sumset. □

Lemma 4.3 (Cheapest mask of given peak)**.** *For* $h \in (0, 1]$ *let* $r \geq 0$ *satisfy* $1/(2r+3) < h \leq 1/(2r+1)$ *and* $\rho = 1 - (2r+1)h \in [0, 2h)$*. Let* q_h *place mass* h *on each of* $\{-r, \ldots, r\}$ *and* $\rho/2$ *on each of* $\pm(r+1)$*. Then* $\mathbb{E}_q|Y| \geq C(h) := \mathbb{E}_{q_h}|Y| = (r+1)(1-(r+1)h)$ *for every* q *with* $\|q\|_\infty \leq h$*.*

Proof. Direct computation gives the cost formula. Let $R = r + 1$ and $\Delta = q - q_h$. For $|y| < R$ both factors $|y| - R \leq 0$ and $\Delta(y) \leq 0$, so their product is nonnegative. For $|y| = R$ the term vanishes. For $|y| > R$ both factors are nonnegative. Hence $\sum(|y| - R)\Delta(y) \geq 0$. □

Theorem 4.4 (Uniform secret)**.** *Let $r \geq 0$ satisfy $r(r+1)/(2r+1) \leq c \leq (r+1)(r+2)/(2r+3)$. Set $h^*(c) = (r+1-c)/(r+1)^2$. Then*

$$\min_{\mathbb{E}|Y| \leq c} V(q) = \frac{1 + (m-1)h^*(c)}{m},$$

attained by $q_{h^*(c)}$.

Proof. Lemmas 4.1 and 4.2 together with $\int_0^h |B_t|\, dt = 1$ give $mV(q) \geq 1 + (m-1)\|q\|_\infty$. The cost constraint and Lemma 4.3 force $\|q\|_\infty \geq h^*(c)$. The candidate has cost exactly c and interval superlevel sets, so equality holds throughout. The argument covers infinite-support competitors. For $m = 1$ or $c = 0$ one has $V \equiv 1$. □

The optimiser is not in general uniform on an interval. Asymptotically, the cost of a height-h filled peak is $C(h) \sim 1/(4h)$ (the continuous water-filling limit), so $h^*(c) \sim 1/(4c)$.

5 Biased bit

Let $\mathbb{P}(X = 0) = p \in (1/2, 1)$, $\mathbb{P}(X = 1) = 1 - p$, and $a = (1-p)/p \in (0, 1)$. Define

$$V_p(q) = \sum_s \max\bigl\{p\, q(s), (1-p)q(s-1)\bigr\}$$

and

$$V_p^*(c) = \inf\{V_p(q) : \mathbb{E}_q|Y| \leq c\}.$$

The baseline is $V(X) = p$. Equivalently,

$$V_p(q) = p + \sum_s \max\bigl(0, (1-p)q(s-1) - p\, q(s)\bigr).$$

For $p = 1/2$ the problem is the case $m = 2$ of Theorem 4.4.

Theorem 5.1 (Zero extra average advantage)**.** *$V_p(q) = p$ if and only if $q(s) \geq a\, q(s-1)$ for every $s \in \mathbb{Z}$. Such laws are precisely the mixtures $q = \sum_L w_L G_L$, where $G_L(s) = (1-a)a^{s-L}\mathbf{1}_{s \geq L}$, $w_L \geq 0$ and $\sum_L w_L = 1$.*

Proof. The first claim is immediate from the excess representation. Given the inequality, set $d(s) = q(s) - a\, q(s-1) \geq 0$. Iterating and using $a^n q(s-n) \to 0$ yields $q(s) = \sum_{L \leq s} d(L)a^{s-L}$. Summing over s gives $\sum d(L) = 1 - a$, so the weights $w_L = d(L)/(1-a)$ produce the claimed mixture. Conversely each G_L satisfies the inequality, which is preserved by mixtures. □

Corollary 5.2. *No finite-support mask achieves $V_p(q) = p$. If $q(s_0) > 0$ then the ratio inequality forces $q(s_0 + k) \geq a^k q(s_0) > 0$ for all $k \geq 0$.*

Remark 5.3. $V_p(q) = p$ concerns the average only. For a pure geometric the posterior at the left endpoint equals 1, and S is never independent of X.

Theorem 5.4 (Critical cost)**.**

$$c_{\mathrm{crit}}(p) = \min_{\ell \in \mathbb{Z}_{\geq 0}} \Big(\ell + \frac{a(2a^{\ell} - 1)}{1 - a} \Big),$$

attained at $\ell^* = \#\{j \geq 1 : a^j > 1/2\} = \lceil \log 2 / \log(1/a) \rceil - 1$ *(an integer median; with a tie when* $a^{\ell^*+1} = 1/2$*). Moreover* $V_p^*(c) = p$ *for* $c \geq c_{\mathrm{crit}}(p)$ *and* $V_p^*(c) > p$ *for* $c < c_{\mathrm{crit}}(p)$*.*

Proof. By Theorem 5.1 and linearity the cost of a zero-advantage mask is at least the minimum over pure shifts. For positive shifts the cost is larger, so it suffices to minimise $f(\ell) = \mathbb{E}|G - \ell|$ over $\ell \geq 0$. The difference $f(\ell+1) - f(\ell) = 1 - 2a^{\ell+1}$ changes sign at the stated integer, with a tie exactly when equality holds.

Existence of minimisers: a minimising sequence with $\mathbb{E}|Y| \leq c$ is tight by Markov's inequality. A diagonal argument yields a subsequence converging pointwise to some q. Tightness implies $\sum q = 1$. Fatou's lemma gives $\mathbb{E}_q|Y| \leq c$. Scheffé's lemma then upgrades the convergence to ℓ^1. The bound $|V_p(q) - V_p(q')| \leq \|q - q'\|_1$ gives continuity of the objective, so a minimiser exists. (The case $c = 0$ is covered by Theorem 5.5.) If $V_p^*(c) = p$ for some $c < c_{\mathrm{crit}}(p)$, the minimiser would be a zero-advantage mask of cost strictly below $c_{\mathrm{crit}}(p)$, a contradiction. For $c \geq c_{\mathrm{crit}}(p)$ the geometric at the optimal shift is feasible. □

Theorem 5.5 (Initial segment)**.** *For* $0 \leq c \leq 1 - p$*,* $V_p^*(c) = 1 - pc$*, attained by* $q = (1 - c)\delta_0 + c\delta_1$*. Moreover* $V_p(q) \geq 1 - p\,\mathbb{E}|Y|$ *for every* q*.*

Proof. Choosing the first term of the maximum for $s \leq 0$ and the second for $s \geq 1$,

$$\begin{aligned} V_p(q) &\geq p \Pr(Y \leq 0) + (1 - p) \Pr(Y \geq 0) \\ &= 1 - p \Pr(Y > 0) - (1 - p) \Pr(Y < 0) \\ &\geq 1 - p \Pr(Y \neq 0) \geq 1 - p\,\mathbb{E}|Y|, \end{aligned}$$

using $1 - p < p$ and $\mathbf{1}[y \neq 0] \leq |y|$ on $\mathbb{Z}$. The two-point mask has cost c and vulnerability $1 - pc$ on the stated interval. □

5.1 Two certified intermediate pieces

All lower bounds below come from the following weak-duality device. For any $\alpha : \mathbb{Z} \to [0, 1]$,

$$V_p(q) \geq \sum_y \kappa_\alpha(y) q(y), \qquad \kappa_\alpha(y) := p\alpha_y + (1 - p)(1 - \alpha_{y+1}). \tag{1}$$

If $\kappa_\alpha(y) \geq \mu - \lambda|y|$ for all y with $\lambda \geq 0$, then $V_p(q) \geq \mu - \lambda \mathbb{E}_q|Y| \geq \mu - \lambda c$ for every feasible q.

Theorem 5.6 (Certified cells)**.** *Let* $1/2 < p \le \varphi := (\sqrt{5}-1)/2$*,* $a = (1-p)/p$*, and*

$$c_1 = \frac{1+a^2}{1+a+a^2}, \qquad c_2 = \frac{1+a^2+2a^3}{1+a+a^2+a^3}.$$

Then $1-p < c_1 < c_2$ *and*

$$V_p^*(c) = \begin{cases} p + \dfrac{(1-p)^2}{p}(1-c), & 1-p \le c \le c_1, \\ 1-p+p^3(2-c), & c_1 \le c \le c_2. \end{cases}$$

On the first piece the optimiser is $q(-1) = c - a(1-c)$*,* $q(0) = 1-c$*,* $q(1) = a(1-c)$*. On the second piece, with* $t = p^2(2-c)$*, the optimiser is* $q(-1) = t$*,* $q(0) = at$*,* $q(1) = a^2 t$*,* $q(2) = 1 - t(1+a+a^2)$*. Both affine functions are lower bounds on* $V_p^*(c)$ *for every* $c \ge 0$*.*

Proof. The ordering $1-p < c_1 < c_2$ follows by direct comparison of the rational expressions.

First piece, lower bound. Take $\alpha_y = 1$ for $y \le 0$, $\alpha_1 = (2p-1)/p$, $\alpha_y = 0$ for $y \ge 2$, $\lambda = (1-p)^2/p$, and $\mu = p + \lambda$. Then $\kappa(y) = p$ for $y \le -1$, $\kappa(0) = \mu$, $\kappa(1) = \mu - \lambda$, and $\kappa(y) = 1-p$ for $y \ge 2$. The dual inequality holds with equality at $y = -1, 0, 1$, holds for $y \le -2$, and holds for $y \ge 2$ if and only if $(1-p)^2 \ge p(2p-1)$, i.e., $p \le \varphi$. Thus $V_p^*(c) \ge p + \lambda(1-c)$ for all c.

First piece, attainment. The displayed mask has mass 1, cost c, and is nonnegative precisely when $c \ge 1-p$. The ratio constraint $q(0) \ge a\, q(-1)$ holds precisely when $c \le c_1$. Direct evaluation of the four nonzero terms of V_p ($s = -1, 0, 1, 2$) yields the claimed value.

Second piece, lower bound. Take $\alpha_y = 1$ for $y \le -1$, $\alpha_0 = p(1+p)$, $\alpha_1 = p^2$, $\alpha_y = 0$ for $y \ge 2$, $\lambda = p^3$, and $\mu = 1 - p + 2p^3$. The condition $p \le \varphi$ ensures all weights lie in $[0, 1]$. Direct expansion shows equality of the dual inequality at $y = -1, 0, 1, 2$; the inequality holds for $y \le -2$ and for $y \ge 3$. Thus $V_p^*(c) \ge 1 - p + p^3(2-c)$ for all c.

Second piece, attainment. The displayed mask has mass 1 and cost c. Nonnegativity of the residual holds precisely when $c \ge c_1$; the residual does not exceed the geometric continuation precisely when $c \le c_2$. Evaluation of V_p on the four atoms yields the claimed value. □

Example 5.7. For $p = 3/5$: $a = 2/3$, $c_1 = 13/19$, $c_2 = 11/13$, and $V_p^*(c) = 13/15 - 4c/15$ on $[2/5, 13/19]$. At $c = 1/2$ the optimiser is $(1/6, 1/2, 1/3)$ on $\{-1, 0, 1\}$ and $V_p^* = 11/15$. At $c = 3/4$ one has $V_p^* = 67/100$; at $c = 4/5$ one has $V_p^* = 412/625$.

Remark 5.8 (Range of validity). The bias restriction is genuine. At $p = \varphi$ the two slopes coincide. For $p > \varphi$ the certificates fail. At $p = 3/4$ and $c = 3/10$ the first formula equals $97/120$, yet the feasible mask $q(0) = 51/70$, $q(1) = 17/70$, $q(2) = 2/70$ has cost $3/10$ and vulnerability $223/280 < 97/120$.

Proposition 5.9 (Monotonicity and convexity)**.** *The function* V_p^* *is nonincreasing and convex on* $[0, \infty)$*. It is continuous on* $(0, \infty)$*. Continuity at* 0 *follows from Theorem 5.5.*

Proof. If $c \le c'$ the feasible set for c is contained in that for c', so V_p^* is nonincreasing. For convexity, take feasible masks q_0, q_1 for budgets c_0, c_1 and $\theta \in [0,1]$. The mixture $\theta q_0 + (1-\theta)q_1$ is feasible for $\theta c_0 + (1-\theta)c_1$ and, by convexity of V_p, has vulnerability at most the corresponding convex combination. Taking infima yields convexity of V_p^*. A finite convex function on an open interval is continuous, so V_p^* is continuous on $(0, \infty)$. □

Remark 5.10 (Counterexample to a uniform geometric-ratio claim). At $p = 3/5$ and $c = 1/2$ the optimal mask on $\{-1, 0, 1\}$ is $(1/6, 1/2, 1/3)$. Here $a = 2/3$, but $q(0)/q(-1) = 3 \neq a$. Thus an optimal intermediate mask need not have geometric ratios on its entire support.

Open question 5.11. *What is the structure of an optimal mask for a biased bit when $1 - p < c < c_{\text{crit}}(p)$? The two certified families of Theorem 5.6 show that a free residual may appear at either endpoint. Whether every intermediate optimum is residual-adjusted (with a free residual at exactly one endpoint) remains open. No claim is made about the number of pieces.*

Figure 1 displays the cost–vulnerability frontier. Solid curves are the proved segments (initial piece, the two certified intermediate pieces for $p \le (\sqrt{5}-1)/2$, and the baseline). Dashed curves are finite-support LP upper bounds only and are not certified.

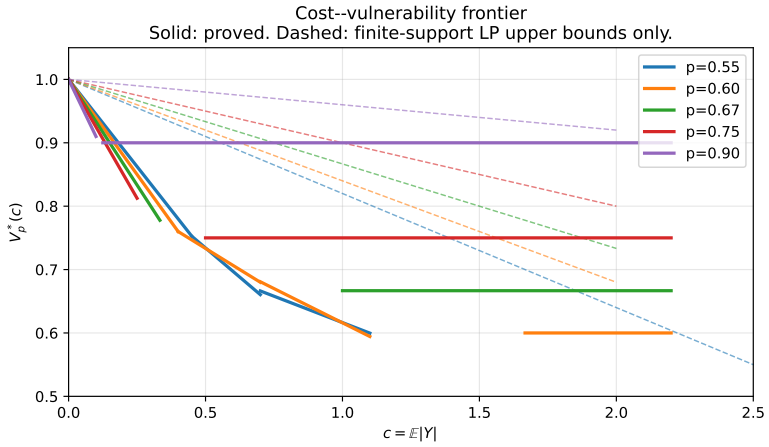


Figure 1: Cost–vulnerability frontier. Solid: proved segments. Dashed: finite-support LP upper bounds only. The dashed curves are not certified.

6 Limitations and future work

The model is elementary: one observation, one guess, independence, and a pure absolute-moment cost. The frontier between c_2 and $c_{\text{crit}}(p)$ for $p \le \varphi$, and

Table 1: Values of $V_p^*(c)$. Exact entries are proved. Entries marked $\approx$ are rounded objective values of finite-support linear programs on $\{-40, \dots, 40\}$ and are upper bounds only.

c	0	0.25	0.5	0.75	1	1.5
$p = 1/2$ (uniform $m = 2$)	1	7/8	3/4	21/32	5/8	7/12
$p = 0.6$	1	17/20	11/15	67/100	≈ 0.6314	≈ 0.6036
$p = 2/3$	1	5/6	≈ 0.7333	≈ 0.6871	2/3	2/3
$p = 3/4$	1	13/16	3/4	3/4	3/4	3/4
$p = 0.9$	1	0.9	0.9	0.9	0.9	0.9

between $1 - p$ and $c_{\mathrm{crit}}(p)$ for $p > \varphi$, remains open. Extensions to nonuniform multi-valued priors, other costs, modular arithmetic, or multiple guesses are likewise open.

7 Conclusion

The optimal additive integer masks have been determined for a uniform interval secret and, for a biased bit, for the zero-advantage regime, the critical cost, the initial segment, and two further certified pieces when the bias does not exceed the golden-ratio conjugate. The results supply exact design guidance for the average one-guess metric inside a transparent model. They do not by themselves yield a concrete cryptographic masking scheme, and zero average guessing advantage is not perfect secrecy.

Acknowledgment

Portions of the drafting, proof structuring, figure generation, verification scripts, and release packaging were assisted by AI tools (Grok and Grok Bot). The author checked all results and remains fully responsible for the accuracy of all statements, proofs, and numerical results.

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